

Conductor Descent for Prime-Power Lights Out: Further Attempts and Their Exact Limitations

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Abstract

This addendum investigates the general identity $d(p^k - 1) = d(p - 1)$ rather than additional prime-specific zero-nullity families. It does not prove the general identity or give a counterexample. We prove a precise obstruction to descent by powering in the regular-map operations, an exclusion of one projected mixed-collision phase, and an exact cyclotomic norm congruence. Each result is separated from the stronger, still unproved assertion that would be needed for conductor descent.

1 The target

Let $\mu_N \subset \overline{\mathbb{F}}_2^\times$ denote the N th roots of unity. For $p > 5$, the outstanding assertion is that

$$1 + u + u^{-1} + v + v^{-1} = 0, \quad u, v \in \mu_{p^j}, \quad (1)$$

forces both coordinate orders to divide p . The known proof reduces the problem to $j \leq c_p = v_p(2^{\text{ord}_p(2)} - 1)$. The companion note [1] covers the additional order classes $3 \mid \text{ord}_p(2)$ and $v_2(\text{ord}_p(2)) = 2$. Neither reduction disposes of all possible Wieferich primes.

The substitution

$$u = xy, \quad v = x/y$$

is invertible on $\mu_{p^j}^2$, because its exponent matrix has determinant -2 . It converts (1) into

$$(x + x^{-1})(y + y^{-1}) = 1. \quad (2)$$

It preserves the maximum coordinate order, but need not preserve the individual orders. In particular, “mixed” in the additive coordinates and “mixed” in the product coordinates are not interchangeable notions.

2 The group-theoretic route

Proposition 2.1 (Exact group formulation). *For $p > 5$, a higher-level solution is equivalent to a configuration in some $\text{SL}_2(\mathbb{F}_{2^h}) = \text{PSL}_2(\mathbb{F}_{2^h})$ with*

$$|B| = 2, \quad |A^2B| = 3, \quad |A|, |AB| \text{ powers of } p, \quad \max(|A|, |AB|) \geq p^2.$$

Proof. Given (2), put $a = x + x^{-1}$. Then $a \neq 0, 1$: the exceptional values would give x order 1 or 3. Set

$$A = \begin{pmatrix} a & 1 \\ 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & a + a^{-1} \\ 0 & 1 \end{pmatrix}.$$

Both determinants are one, B is a nonidentity involution, and

$$\mathrm{tr}(A) = a, \quad \mathrm{tr}(AB) = a^{-1}.$$

The eigenvalues of A are x, x^{-1} and those of AB are y, y^{-1} , so their orders are exactly the coordinate orders. Cayley–Hamilton gives

$$A^2 = aA + I, \quad \mathrm{tr}(A^2B) = a \mathrm{tr}(AB) + \mathrm{tr}(B) = 1.$$

A determinant-one matrix of trace one in characteristic two has characteristic polynomial $T^2 + T + 1$ and exact order three.

Conversely, the prescribed orders give $\mathrm{tr}(B) = 0$ and $\mathrm{tr}(A^2B) = 1$. Cayley–Hamilton therefore implies $\mathrm{tr}(A) \mathrm{tr}(AB) = 1$. The eigenvalues of the two odd-order matrices recover (2), with the required maximum coordinate order. $\square$

Thus the trace-product formulation is not irrelevant to the original trace-sum problem. The coordinate change must be taken into account when interpreting the discussion in [2].

The generating pair describes an orientably regular map with face length $|AB|$ and vertex valency $|A|$. For $\gcd(j, |A|) = 1$, Wilson’s hole operation H_j replaces (A, B) by (A^j, B) ; its face length is $|A^jB|$ [3]. The next proposition tests the most direct proposed descent. It also allows noninvertible exponents, which are not the usual Wilson operations but could otherwise have reduced the vertex order.

Proposition 2.2 (No locus-preserving powering descent). *Let $n = |A|$ be odd and let A, B be as in the preceding construction. For every positive integer j ,*

$$|A^{2j}B| = 3 \iff j \equiv \pm 1 \pmod{n}.$$

Consequently a powering step that retains the defining order-three condition preserves both the vertex and face orders.

Proof. Let $D_0(X) = 0$, $D_1(X) = X$ and $D_{j+1}(X) = XD_j(X) + D_{j-1}(X)$ over $\mathbb{F}_2$. The trace recurrence and its initial values give

$$\mathrm{tr}(A^jB) = \frac{D_j(a)}{a^2}.$$

Since $D_{2j}(a) = D_j(a)^2$,

$$|A^{2j}B| = 3 \iff D_j(a)^2 = a^2 \iff D_j(a) = a.$$

Writing the eigenvalues of A as x, x^{-1} , the last equality is $x^j + x^{-j} = x + x^{-1}$. The quadratic $T^2 + aT + 1$ has exactly the roots x, x^{-1} . It follows that $x^j = x$ or x^{-1} , equivalently $j \equiv 1$ or $-1 \pmod{n}$. In this case $\mathrm{tr}(A^jB) = a^{-1}$ as well, so the face order is unchanged. $\square$

Replacing (A, B) by (AB, B) swaps the two odd orders and retains the trace-product condition. Group automorphisms preserve orders. Therefore a sequence of these operations whose intermediate configurations all retain the order-three condition cannot perform conductor descent. This does not rule out a sequence which leaves that locus and returns using an additional argument.

There is also an entirely group-theoretic explanation for a tempting but unhelpful observation. Put $R = A^2B$. From $B^2 = R^3 = 1$,

$$A^4B = RA^{-2}R^{-1}.$$

Indeed, with $U = A^2 = RB$, $RU^{-1}R^{-1} = RBR = UR = U^2B$. Thus $|A^4B| = |A|$ for every odd-order A . The equal-order map obtained after two H_2 operations is automatic; it does not impose a restriction on prime-power exponents.

3 A lower-algebra test for mixed collisions

Write $F_0 = 0$, $F_1 = 1$, $F_{n+1} = XF_n + F_{n-1}$, and, for odd n ,

$$H_n = F_{(n-1)/2} + F_{(n+1)/2}.$$

Then $D_n = XH_n^2$, and H_n is square-free with roots the nonzero traces of nonidentity n th roots of unity. Fix $p > 3$, put $m = (p-1)/2$, and write

$$H = H_p, \quad H^*(X) = X^m H(1/X), \quad R(X) = D_p(1/X).$$

Denominators in the following statements are invertible modulo H , because $H(0) = 1$.

Lemma 3.1 (Exact reciprocal mixed criterion). *For β a root of H , the reciprocal β^{-1} is the trace of a primitive p^2 th root if and only if*

$$H(R(\beta)) = 0.$$

Proof. Choose u satisfying $u + u^{-1} = \beta^{-1}$. Then $u^p + u^{-p} = R(\beta)$. If the latter is a root of H , the two solutions of its reciprocal quadratic are primitive p th roots, so u^p has exact order p . Hence u has exact order p^2 . The converse follows from the same trace identity. In particular $R(\beta) = 0$ is an old collision, not a mixed one: $H(0) = 1$ already excludes it from this criterion. $\square$

Proposition 3.2 (Exclusion of the fixed projected phase). *Define*

$$Q_p(X) = H^*(X) + X^{m+1} + XH(X).$$

Then

$$\gcd(H, R - X) = \gcd(H, Q_p), \quad \deg Q_p < m, \quad Q_p(0) = 1.$$

In particular, if H_p is irreducible, then $D_p(\beta^{-1}) = \beta$ is impossible for a root β of H_p .

Proof. The Dickson identity gives

$$R(X) - X = X^{-p} (H^*(X) + X^{m+1})^2.$$

Clearing the invertible denominator and removing the square does not change the gcd with the square-free polynomial H . Adding XH does not change it either.

The two highest coefficients of H are 1, 1, and $H(0) = 1$. Thus the degree- $m+1$ and degree- m terms of Q_p cancel, while its constant term is one. An irreducible degree- m polynomial cannot share a root with this nonzero lower-degree polynomial. $\square$

This is an exclusion of one branch, not of all mixed collisions. If H is irreducible and one mixed collision exists, Frobenius equivariance and transitivity imply

$$R(\beta) = \beta^{2^k}$$

for a fixed phase k on the entire root orbit. Proposition 3.2 excludes $k = 0$ only. For every odd a ,

$$\gcd(H, R - D_a) = \gcd(H, H^* + X^{m+1}H_a),$$

by the same squared-numerator argument. Choosing an odd representative $a \equiv \pm 2^k \pmod{p}$ reduces the other phases to explicit polynomials, but does not prove their coprimality. If H is reducible, even the same-orbit phase assumption is unjustified: one lower orbit may map to another.

Proposition 3.3 (Only one candidate phase when H_p is irreducible). *Assume $H = H_p$ is irreducible. Define*

$$\mathcal{Q} = \frac{H^*}{H} \pmod{X^{m+1}}, \quad t = \deg \mathcal{Q}, \quad \mathcal{B} = \frac{H^* + \mathcal{Q}H}{X^{m+1}},$$

where the representative $\mathcal{Q}$ has degree at most m . Then a reciprocal mixed collision of orders p^2, p exists if and only if

$$t \geq 1 \quad \text{and} \quad \mathcal{B} = H_{2t-1}.$$

Proof. The truncated inverse exists because $H(0) = 1$, and the defining congruence makes $\mathcal{B}$ a polynomial. Since H is irreducible, its roots form one Frobenius orbit of size m . The signed powers of two therefore exhaust the nonzero residue classes modulo p up to sign. By Lemma 3.1, a mixed collision is equivalent to $R(\beta) = D_a(\beta)$ for some odd a with $1 \leq a \leq p-2$. The preceding squared-numerator identity makes this equivalent to

$$H^* + X^{m+1}H_a = CH$$

for a polynomial C . Comparing leading degrees gives $\deg C = (a+1)/2 \leq m$. Reducing modulo X^{m+1} then gives $C = \mathcal{Q}$. It follows that $a = 2t-1$ and $\mathcal{B} = H_a$.

Conversely, the displayed equality with $t \geq 1$ supplies such an odd $a = 2t-1 \leq p-2$, so R maps a root of H to a root of H . Lemma 3.1 supplies the mixed collision. If $t = 0$, no quotient C of the required positive degree exists. $\square$

This reduces the possible phases to one explicit identity in the lower polynomial algebra. It is not a proof that the identity always fails. For example, at $p = 11$,

$$H = X^5 + X^4 + X^2 + X + 1, \quad \mathcal{Q} = X^5 + X^4 + X^2 + 1, \quad \mathcal{B} = X^4 + X^2 + 1 \neq H_9.$$

Simple coefficient tests do not uniformly decide the identity: at $p = 1931$, one gets $t = 964$, and $\mathcal{B}$ agrees with H_{1927} in degrees zero through eight before differing in degree nine. No argument excluding the equality for arbitrary p was found.

Remark 3.4 (A countermodel to dropping prime support). *Reciprocal mixed descent is false for general odd periods. The polynomial $H_7 = X^3 + X^2 + 1$ has reciprocal $X^3 + X + 1$, and*

$$H_9 = (X+1)(X^3 + X + 1).$$

Thus the reciprocal of a primitive 7-trace is a primitive 9-trace. Both orders divide 21^2 , while 9 does not divide 21. This is not a prime-power counterexample; it shows that an argument using only oddness and $M \mid M^2$ cannot suffice.

Remark 3.5 (A failed auxiliary cubic obstruction). *For the remaining irreducible cases with $\text{ord}_p(2) = p-1$, $m = (p-1)/2$ odd and $3 \nmid m$, one might hope that every reciprocal lift of a p -trace has order divisible by three. That would exclude a p -primary lift, but the auxiliary assertion is false.*

Take $p = 107$ and

$$K = \mathbb{F}_2[z]/(1 + z + \cdots + z^{106}), \quad \beta = z + z^{-1}.$$

Encode an element by placing its coefficient of z^i in bit i . The element

$$u = 0x21500a7574a334aa348c8366ce2$$

then satisfies, by exact polynomial reduction,

$$(u + u^{-1})\beta = 1, \quad \text{ord}(u) = 3002399751580331 = 107 \cdot 28059810762433.$$

This order is not divisible by three. It is also not a power of 107, so this example refutes only the auxiliary assertion, not the Lights Out conjecture.

4 Exact local valuations do not force global descent

Lemma 4.1 (Simple two-adic intersection). *Let u, v satisfy (1), and let U, V be their Teichmüller lifts in an unramified extension of $\mathbb{Q}_2$. Put*

$$S = 1 + U + U^{-1} + V + V^{-1}.$$

If the coordinate orders are powers of $p > 5$, then $v_2(S) = 1$.

Proof. Let σ be the unramified Frobenius, and put $A = U + U^{-1}$, $B = V + V^{-1}$. Direct expansion gives

$$S^2 - \sigma(S) = 2(S + 1 + AB).$$

Write $S = 2W$, as its reduction vanishes. Dividing by two and reducing modulo two yields

$$\overline{W}^2 = 1 + \alpha\beta = 1 + \alpha + \alpha^2, \quad \alpha = u + u^{-1}, \quad \beta = \alpha + 1.$$

If the right side vanished, multiplying by u^2 would give $u^4 + u^3 + u^2 + u + 1 = 0$, so u would have order 5. Thus $\overline{W} \neq 0$ and $v_2(S) = 1$. $\square$

Proposition 4.2 (Norm congruence). *Let $N = p^j$, $p > 5$, $L = \mathbb{Q}(\zeta_N + \zeta_N^{-1})$, and*

$$S = 1 + \zeta_N^a + \zeta_N^{-a} + \zeta_N^b + \zeta_N^{-b}, \quad M = N_{L/\mathbb{Q}}(S).$$

Then

$$M \equiv \left(\frac{5}{p}\right) \pmod{p^j}. \quad (3)$$

If $h = \min\{r > 0 : 2^r \equiv \pm 1 \pmod{p^j}\}$ and exactly t primes of L above two divide S , then

$$v_2(M) = ht.$$

Proof. At the unique prime above p , every conjugate of S is congruent to 5, so $p \nmid M$ and

$$M \equiv 5^{[L:\mathbb{Q}]} \equiv (5/p) \pmod{p}.$$

For every rational prime $\ell \neq p$, the residue degree e of a prime of L above ℓ is the least positive integer with $\ell^e \equiv \pm 1 \pmod{N}$. Hence the norm of each prime-ideal factor of (S) is ± 1 modulo N . Their product gives $M \equiv \pm 1 \pmod{N}$, with its sign uniquely selected by the congruence modulo p . This proves (3). The second assertion follows from Lemma 4.1 at every prime above two and the prime-ideal norm formula. $\square$

On a Wieferich plateau, $2^h \equiv 1$ or $-1 \pmod{p^j}$. Therefore the exact two-part 2^{ht} is already compatible with the norm congruence. The odd part and the sign of M are not controlled by these statements.

Nor do arbitrary global congruences repair this issue. For any prescribed subset of the primes above two, the Chinese remainder theorem produces an algebraic integer divisible simply at exactly those primes, while agreeing with S to any prescribed finite p -adic precision. Such an integer need not have the special five-term shape or its small height. Thus this observation is not a counterexample; it pinpoints the information missing from an argument using only local valuation and norm congruences.

5 What this investigation has and has not established

The direct locus-preserving powering descent cannot work. The global norm calculation is exact but consistent with a higher conductor. Under an irreducibility hypothesis, the mixed-collision argument eliminates the fixed projected phase and reduces all remaining phases to one explicit polynomial identity. It does not prove that the identity fails, or handle collisions between distinct lower orbits.

Consequently none of these three attempts proves the all-prime identity, and no prime-power counterexample has been obtained. In particular, the equivalent group formulation must not be promoted to a group-theoretic exponent bound without an additional argument.

References

- [1] *Two Trace–Norm Theorems for Prime-Power Lights Out Stabilization*, `tex\prime_power_endpoint\prime_power_endpoint.tex`.
- [2] W. Boyles, *Reciprocal Five-Term Relations among p^j -th Roots of Unity in Characteristic Two: Reductions, Obstructions, and the Wieferich Plateau*, `tex\five_term_plateau\five_term_plateau.tex`.
- [3] G. Gévay and G. A. Jones, *Hole operations on Hurwitz maps*, Sections 2 and 6, <https://arxiv.org/abs/2111.05566>.